📘 Question 1: Type I and Type II Errors

- **Type I error**: Rejecting the null hypothesis when it is actually true (false positive).
- **Type II error**: Failing to reject the null hypothesis when it is false (false negative).
- **Impact**: Type I errors lead to incorrect conclusions of significance, while Type II errors cause missed discoveries.

📘 Question 2: P-value

- **P-value**: Probability of obtaining results at least as extreme as the observed data, assuming the null hypothesis is true.
- **Interpretation**:
    - Small p-value (< 0.05): Evidence against null hypothesis.
    - Large p-value (> 0.05): Fail to reject null hypothesis.

📘 Question 3: Z-test vs T-test

- **Z-test**: Used when population variance is known and sample size is large (n > 30).
- **T-test**: Used when population variance is unknown and sample size is small.
- **Difference**: Z-test relies on normal distribution; T-test uses Student's t-distribution.

📘 Question 4: Confidence Interval

- **Confidence interval**: Range of values within which the true population parameter is expected to lie.
- **Margin of error**: Larger margin → wider interval, less precision. Smaller margin → narrower interval, more precision.

📘 Question 5: ANOVA Test

- **ANOVA**: Compares means across 3+ groups.
- **Assumptions**: Normality, independence, equal variances.
- **Purpose**: Extends hypothesis testing beyond two groups.

📘 Question 6: Python Program – One-Sample Z-test

```
import numpy as np
from statsmodels.stats.weightstats import ztest

data = np.array([5.1, 5.3, 5.0, 5.2, 5.4, 5.1])
z_stat, p_val = ztest(data, value=5.0)

print("Z-statistic:", z_stat)
print("P-value:", p_val)
```

- **Interpretation**: If p-value < 0.05, reject null hypothesis.

📘 Question 7: Binomial Distribution Simulation

```
import numpy as np
import matplotlib.pyplot as plt

data = np.random.binomial(n=10, p=0.5, size=1000)
plt.hist(data, bins=11, edgecolor='black')
plt.title("Binomial Distribution (n=10, p=0.5)")
plt.show()
```

📘 Question 8: Central Limit Theorem

```
import numpy as np
import matplotlib.pyplot as plt

samples = [np.mean(np.random.exponential(scale=2, size=100)) for _ in range(1000)]
plt.hist(samples, bins=30, edgecolor='black')
plt.title("Central Limit Theorem Demonstration")
plt.show()
```

- **Result**: Sample means approximate normal distribution.

📘 Question 9: Confidence Interval Visualization

```
import numpy as np
```

```
import matplotlib.pyplot as plt
import scipy.stats as st

data = np.random.randn(50)
mean = np.mean(data)
conf_int = st.t.interval(0.95, len(data)-1, loc=mean, scale=st.sem(data))

plt.hist(data, bins=10, edgecolor='black')
plt.axvline(conf_int[0], color='red', linestyle='--')
plt.axvline(conf_int[1], color='red', linestyle='--')
plt.title("95% Confidence Interval")
plt.show()

print("Confidence Interval:", conf_int)
```

📘 Question 10: Chi-square Goodness-of-Fit

```
import numpy as np
from scipy.stats import chisquare

observed = np.array([50, 30, 20])
expected = np.array([40, 40, 20])

chi_stat, p_val = chisquare(f_obs=observed, f_exp=expected)
print("Chi-square statistic:", chi_stat)
print("P-value:", p_val)
```

- **Interpretation**: If p-value < 0.05, observed distribution significantly differs from expected.
